

Intelligent Operations with **SWARM AVA**

A practical guide to intelligent decision-making with Agentic AI based optimization

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Introduction

SWARM AVA is a digital agent designed to act as a partner and operational consultant, uncovering valuable insights across your business functions and processes. Built as a superset of OpenAI ChatGPT, AVA includes extended capabilities tailored specifically to improve the operations of agrifood businesses.

AVA accelerates access to organizational knowledge, helps solve challenges, and continuously learns from corporate and tribal knowledge to deliver actionable insights and ongoing improvements. Ultimately, the goal is to increase the efficiency and effectiveness of the business, and of the people within it.

This document outlines key features already available, and those expected to come online during 2025. Some capabilities (e.g. conversational voice) are reliant on OpenAI API features being made available. Many other features are in development by SWARM directly. As a superset of OpenAI, new capabilities added to the API will also be available within AVA.

While AVA is a standalone product, it can also integrate tightly with SWARM ENGINE, providing a way to generate and interact with AI applications to solve complex operational problems to reduce costs and improve margins. We envision a future where **every decision made by operations teams** is optimized with AI in real-time.



By the **end of 2026**, Agentic AI will feel like a **thinking entity** that acts as a true **consultant, researcher, strategist, and exec assistant**.

- **Autonomous:** acts independently, with user guidance
- **Proactive:** anticipates needs, does not simply react
- **Self-improving:** learns, and adapts behavior
- **Highly integrated:** across software, physical, and cloud systems

Companies deploying Agentic AI will have a strategic advantage over those that adopt a 'wait and see' approach

The Problem

Modern operations teams face a daily avalanche of data, decisions, and disruptions – yet the tools they rely on to interpret and act on this information are often fragmented, reactive, and slow. Teams spend hours searching for answers, double-checking reports, and manually coordinating across systems that were never designed to work together. The result: missed opportunities, costly errors, and an overwhelming cognitive load. Common pain points in this scenario include:

- **Too much data, too little insight**

Operational systems generate terabytes of data – but teams spend hours digging through dashboards, emails, spreadsheets, and ERPs just to answer simple questions. Valuable insights are buried, delayed, or lost.

- **Decisions depend on who's available**

Critical business knowledge often lives in people's heads. When they're out, on leave, or overloaded, the team slows down or repeats past mistakes.

- **Siloed systems and conversations**

Sales, ops, logistics, quality – each team uses different tools, with little integration. Conversations are scattered across slack, email, and meetings, with no single source of truth.

- **Firefighting is standard practice**

Most alerts come too late to act. Teams scramble to fix issues after the fact – whether it's a missed delivery, a quality breach, or a labor gap. Strategic thinking gets pushed aside by daily chaos.

- **New hires take too long to ramp**

Without access to institutional memory or clear, contextual answers, new employees take weeks or months to become fully productive. The cost of turnover is amplified by knowledge gaps.

A Solution

SWARM AVA is a digital agent that exists to solve these issues, acting as an intelligent partner that proactively surfaces insights, learns from tribal knowledge, and connects people, data, applications, agents, and solutions so that decisions can be made rapidly and accurately.

"I get answers in seconds that used to take me an hour to answer"

"It's like having our best analyst available 24/7, and we never get put on their backlog"

"We didn't realize how siloed we had become until AVA started piecing the information together"

"We used to spend the morning reacting to yesterday's problems. Now, we get advance warning and avoid issues"

"Our new planner was up to speed in weeks, and it used to take months before"



Implementation Process

Getting started with SWARM AVA is fast, and low risk. AVA is designed to integrate seamlessly into your existing workflows – whether that means answering operational questions, uncovering inefficiencies, generating reports, or guiding users through challenge discovery. Most organizations begin seeing value in days, not months. Rather than requiring a major rollout or IT overhaul, AVA can be deployed gradually, starting with a single team, function, or problem, and scaled as confidence and familiarity grow. We typically follow a process like this:

1

Define the use cases that matter

You don't need to boil the ocean. We'll help you identify the highest-value scenarios – whether that's surfacing insights from your ERP, guiding new hires, or automating alerts. Our team can facilitate short sessions to frame key problems.

2

Deploy your AVA instance

SWARM provides a secure, SOC2-compliant AVA instance tailored to your environment. Access is controlled through the Administration Console, where you can define users, data permissions, recommendations, and integrations.

3

Connect your data

AVA works immediately with public and user-provided content. To unlock deeper insights, you can choose to connect AVA to internal systems like Box, Confluence, SharePoint, ERP datasets, or SQL tables. You control what AVA can access.

4

Start the conversation

Users simply ask questions – with minimal training required. AVA explains documents, traces logic, interviews stakeholders, and provides decision support through natural conversation.

5

Learn, adapt, scale

AVA continually improves. It captures patterns and refines recommendations. As new use cases emerge, you can activate extended capabilities – such as proactive alerts, workflow automation, or integration with the SWARM ENGINE for full optimization loops.

Why it's easy and risk-free:

- ✓ No rip-and-replace of existing systems
- ✓ Works out-of-the-box with conversational input
- ✓ Scales from one user to enterprise-wide deployments
- ✓ Transparent, explainable recommendations
- ✓ Enterprise-grade security and compliance (SOC 2)
- ✓ Supported by our expert data science and customer success teams



Human vs. Agentic AI Decision-Making

Traditional decision-making relies heavily on human judgment, spreadsheets, and static rules that are often constrained by time, siloed data, and limited visibility into the full operational landscape. While human experience and knowledge is critical, it struggles to scale or adapt in complex, disruptive, or high-velocity environments.

Agentic AI, as embodied by SWARM AVA and the SWARM ENGINE, introduces a new paradigm: teams of agents that continuously analyze, negotiate, and optimize decisions in real time across thousands of variables, constraints, and outcomes. This enables faster, more precise, and more transparent decisions at every level of the organization. Agents can also be trained to specialize in certain tasks, such as interviewing a user about a supply chain challenge, building a Machine Learning model, retrieving data from corporate datasets, and much more.

	Traditional Decision-Making	Agentic AI
Speed	Slow, time-consuming analysis	Real-time or near-real-time decisions
Scalability	Limited by team size / capacity	Scales easily across sites, SKUs, etc.
Data Usage	Often siloed or simplified	Can integrate diverse data sources
Transparency	Decisions based on gut or opaque (tribal) logic	Clear rationale and traceable logic for every decision
Explore Scenarios	Difficult, time-intensive	Easy to simulate millions+ “what-if” scenarios to find optimal solutions
Adaptability	Requires manual updates to processes; disruption is hard	Learns and adapts to new constraints or conditions automatically
Consistency	Varies by person, team, or day	Consistent decisions aligned to defined objectives
Cost of Errors	High! Manual errors often go unnoticed	Much lower. AI flags anomalies, explores optimal outcomes
Talent Dependency	Relies heavily on individuals, can be hard with vacations or exits	Augments teams, reduces risk of single-point knowledge



Why now?

We're at a turning point in how work gets done. The complexity of modern operations is outpacing the capacity of traditional tools, dashboards, and human memory. Meanwhile, Agentic AI – capable of acting independently, understanding nuance, and learning from context – has matured from a research concept into a real operational partner.

The speed of business has outpaced human bandwidth

With thousands of signals to monitor and decisions to make daily, operations teams are constantly overloaded. AVA acts as a tireless digital assistant that never forgets, never sleeps, and works across departments on your behalf, under your direction.

Knowledge is walking out the door

Retirements, resignations, and organizational churn mean critical expertise is disappearing. AVA preserves and scales tribal knowledge, turning what a few key people understand and recall into true institutional capability.

Data is abundant – but underused

Despite enormous investment in systems and analytics, most operational data still goes unread, uncorrelated, or misunderstood. This is especially true for unstructured, document-based data. AVA bridges that gap, surfacing insights and suggesting actions – without needing a data science degree.

AI is no longer experimental

Generative and predictive AI models have proven their value in personal productivity. AVA makes advanced AI practical and usable for everyday enterprise decisions, through a familiar, secure conversational interface.

Decisions need to be faster, clearer, and more resilient

In volatile times – with supply chain shocks, labor shortages, climate disruptions, and shifting markets – the cost of slow or misaligned decisions grows fast. AVA helps teams anticipate risk, adapt quickly, and align actions to strategic goals.



Companies that embrace intelligent agents will future-proof their operations and gain competitive advantage. The laggards may struggle to catch up and will be left behind.

ILLUSTRATIVE EXAMPLES



Case Study #1 – Eliminating losses in transload operations

When Maria Lopez, Regional Logistics Manager at Madanzo, started reviewing AVA's weekly summary reports, she wasn't expecting anything groundbreaking. "Honestly, I thought it would just confirm what we already knew," she said. "Our transload site in Iowa had been hitting its rail targets. Everything looked fine."

But in week two, AVA flagged something odd.

"There's a consistent drop in average load weight per railcar at the Council Bluffs site – down 8% compared to the same period last year," AVA reported.

Maria raised an eyebrow. Railcars were moving on time. Volume was steady. Why the weight discrepancy?

Investigation with AVA

Maria asked AVA to dig deeper.

"What's causing the drop in railcar load weight?"

AVA cross-referenced truck arrival data, driver logs, and loading bay schedules. Within seconds, AVA had an answer:

"Frequent delays in inbound truck deliveries during the 1-4pm window are leaving loading bays idle for up to 40 minutes. This is leading to underfilled railcars being dispatched to avoid penalties."

That was it – the missing piece. Maria brought it to the daily ops team standup.

"We're losing revenue and rail capacity because of avoidable gaps in truck scheduling," she explained to the team. "Let's realign those delivery slots, especially for feed grain shipments on Monday and Wednesday afternoons."

The Turnaround

Within four weeks, the rescheduled truck routes reduced dead time at the bay. AVA monitored the changes and confirmed the improvements in near real-time.

"Average load weight is now within 1% of historical norms. No further idle periods detected in the 1-4pm window."

By the end of the second month, demurrage charges had dropped by **22%**, and the site's overall throughput had improved by **12%**.

Reflection

"What blew me away wasn't just the insight," Maria said. "It was how fast AVA spotted something none of us were looking for. It wasn't in a KPI dashboard. It was buried in the rhythm of our day-to-day."

AVA remains part of Maria's daily routine – listening, watching, and surfacing quiet inefficiencies before they become costly habits.



Case Study #2 – Preventing a feed recall with real-time alerts

As Quality Assurance Director at Madanzo, Darren Finch had seen what a contaminated feed lot could do to customer trust – and to the bottom line. “One bad batch, and you’re dealing with a recall, lost product, and maybe a lost client,” he said. “You can’t afford to miss an alert.”

That’s why Darren had asked AVA to keep watch on USDA updates for anything that might affect incoming corn shipments.

The Alert

On a quiet Thursday morning in June, AVA chimed in:

“A USDA bulletin has reported elevated aflatoxin levels in corn sourced from Central Texas.”

Darren raised his head from his desk.

“AVA, do we have any loads in transit from that region?”

AVA responded instantly:

“Yes. Three inbound shipments scheduled for delivery tomorrow originated within the affected counties. Estimated value: \$147,000. Would you like to initiate secondary testing and hold release to processing?”

Darren didn’t hesitate. He clicked “Yes.”

Swift Action

Those three loads were flagged in the internal ERP system and redirected for lab testing. The QA team ran toxin screens that afternoon. Two shipments passed. One didn’t – showing aflatoxin levels above regulatory thresholds.

“That one would have gone straight into feed if AVA hadn’t flagged it,” Darren said.

The Outcome

The contaminated load was quarantined and later disposed. AVA filed an internal incident summary automatically and updated the QA team’s USDA alert playbook to include similar toxins.

- 0 feed recalls.
- \$500K+ in potential downstream losses avoided.
- Response time reduced from hours to minutes.

Reflection

“This wasn’t about AI replacing anyone,” Darren noted. “It was about having a digital partner watching 24/7 – across sources I don’t have time to check every hour. AVA didn’t just read the alert – AVA understood what it meant, where it hit us, and what we should do next.”

Now, AVA monitors USDA data in real-time – and Darren sleeps a little easier.



Case Study #3 – Turning tribal knowledge into a scalable resource

At Madanzo's headquarters, Rebecca Shaw had been the go-to person for rail logistics for nearly 12 years. She knew every customer preference, the quirks of each terminal, and how to reroute a load on short notice without breaking a sweat. So when Rebecca took sudden medical leave, the team panicked.

"There wasn't a playbook," said Connor, a junior analyst who had just moved over from accounting. "We had files, sure – but none of it told you *why* she made the calls she did."

The Unexpected Backup

Luckily, Rebecca had been using AVA for the past six months – asking questions, generating routing summaries, and reviewing past rate logic. That meant AVA had context. When Connor logged in and asked:

"What was Rebecca's approach to routing grain to the Hutchinson site during winter delays?" AVA responded with a detailed explanation:

"In cases of sub-freezing temps, Rebecca typically prioritized Union Pacific routes over BNSF due to fewer cold-start issues. She also adjusted transit timing to avoid overnight layovers."

Even better, AVA pulled up historical schedules and supplier notes that aligned with that logic. In the weeks that followed, Connor leaned on AVA like a colleague.

"AVA, can you show me what we did the last time a barge was delayed at Catoosa?"

"What's the fallback route for corn going to Tyson when KCS is backed up?"

AVA answered – always with citations and examples, always in plain English.

The Outcome

Shipments stayed on schedule. No escalations. No urgent calls. AVA even flagged two issues Rebecca had manually tracked in her notebook, now automated into alerts.

- Zero disruption in logistics workflows
- Onboarding time cut by 70%
- Team confidence and resilience improved

Reflection

"AVA didn't just help me do Rebecca's job," said Connor. "AVA helped me understand it. That's the difference between surviving a disruption and building something stronger."

When Rebecca returned, she found the department running well, with a trained team member ready to support her with minimal additional training – giving her more time to focus on strategic issues.





CAPABILITIES

Core & Extended Functions

Since the underlying LLM is OpenAI ChatGPT, AVA can offer all of the standard features of this platform, in a secure SOC2 compliant instance, e.g.

- **Natural language conversation:** engage in fluid, human-like dialogue to answer questions, brainstorm ideas, or provide assistance in real time.
- **Context awareness:** retain relevant information about the user and conversation to deliver more personalized and coherent responses.
- **Multilingual support:** converse and translate seamlessly across multiple languages, enabling global collaboration.
- **Content generation:** draft emails, summaries, and other business documents quickly and accurately.
- **Editing and refinement:** improve grammar, tone, structure, or clarity of existing content – from quick fixes to in-depth revisions.
- **Concept and document explanation:** break down complex ideas, explain documentation, or guide users through training material in clear, simple language.
- **Search and summarization:** extract and summarize key information from long documents, emails, PDFs, or knowledge base entries to speed up understanding and reduce manual reading time.
- **Voice interaction [coming soon]:** natural voice-based interaction for hands-free use, pending availability via OpenAI APIs

In addition, AVA has extended capabilities:

- **Operational understanding:** identify and assist with real-world business challenges such as supply/demand balancing, logistics optimization, and inventory management.
- **Interactive report generation:** create high-quality reports on demand, including text analysis, tables, charts, and business summaries.
- **Transparent reasoning:** provide chain-of-thought explanations that show how conclusions were reached and what data was used, improving traceability and trust.
- **Challenge discovery:** interview team members and produce structured Challenge Definition Reports to clarify problems, surface constraints, and quantify business value.
- **Smart recommendations:** suggest relevant people, applications, or agents within your organization who may be able to resolve specific issues.
- **Custom alerts [coming soon]:** notify users when specific thresholds or business rules are met – such as low stock levels or late deliveries.
- **Proactive insights [coming soon]:** surface unexpected anomalies or trends, such as a spike in customer complaints or deviation in trade margins.
- **Enterprise data integration:** Connect securely to internal platforms like AWS, Azure, Postgres, Snowflake [coming soon], Confluence [coming soon], and Box [coming soon], to access structured and unstructured data.



- ## Administration Console

Madanzo

ORGANIZER

MANAGEMENT

Users

AVA Data

AVA Recommender

Home

Console

Nestview

?

People

Apps / Sites

Agents

Category	Department	Name	Email	Phone	Information	Action
Internal	IT	IT help	ithelp@agv.com	(999) 999 - 999	Access to internal IT expert	
Internal	Finance	Financial Advisor	finance@agv.com	(888) 999 - 999	Access to internal financial advice	
Internal	Data Science	Data Analyst	data@agv.com	(888) 888 - 888	Access to internal data analyst	
External	Propero Export Assistance	Blueberry Export Advisor	export@propero.org	(777) 777 - 777	External export advisor specializing in the Peruvian blueberry industry	

Add new person

Description

Facilitates direct access to the company's designated IT specialist, who is equipped to handle technical aspects of the software and hardware systems employed by the company. This node is present to recommend a solution to IT-related issues such as system disruptions, software bugs, or security concerns. It is meant to be recommended to people to solve personal or pressing organizational technical issues. That is, this node is recommended when an IT-related problem, an urgent organizational one, or a personal one, is identified

Tags

Software, hardware, network, IT, technical, system, cybersecurity, database, server, troubleshooting, IT support, error, bug, failure, downtime, outage, slow, performance, upgrade, installation, access, security, CRM, ERP, cloud, SaaS, IoT, AI, machine learning, data analysis, programming, web development

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SWARM

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Examples where AVA can assist

There are many ways that AVA can help out, it really depends on the prompts and imagination of the users. Undoubtedly, the ways in that AVA can assist will also grow over time – as more capabilities are added and as AVA learns more about the users and the organization. Here are some examples, grouped by capability:

1. Answer Questions About the Business

- How many X did we sell last month? Can you group that by salesperson?
- Why is the sales forecast so much lower this month compared to last month?
- What's our current inventory of product Y in region Z?
- Are we on track to meet our Q2 contract obligations to Customer X?
- What's our average margin on bulk soybean meal this month?
- Can you show me the results from Midas, in a tabular format?

The screenshot shows the Madanzo Virtual Advisor interface. On the left, a chat window displays a conversation where the user asks for the latest optimization results from the Midas project, requests a tabular format, and asks for sorting by forecasted volume. The advisor responds with a table of results. On the right, a window titled 'MIDAS SUMMARY RESULTS - April 15th, 2025 09:38am' displays the following table:

Site	Site Details	Inventory
1	28.24	-97.80
2	47.54	-111.05
3	42.70	-84.92
4	40.79	-96.87
5	37.56	-101.15
6	47.67	-101.08
7	33.39	-113.91

Below the table, the chat window shows the advisor's response to the user's request for a tabular format, followed by a SQL query and its results.

SQL Question
What is the total volume from the Benchmark table and the total volume from the Global table?

Query #1
SELECT SUM(CAST(Volume AS FLOAT)) AS Total_Volume_Benchmark FROM madanzo.Benchmark

Query #2
SELECT SUM(CAST(Volume AS FLOAT)) AS Total_Volume_Global FROM madanzo.Global

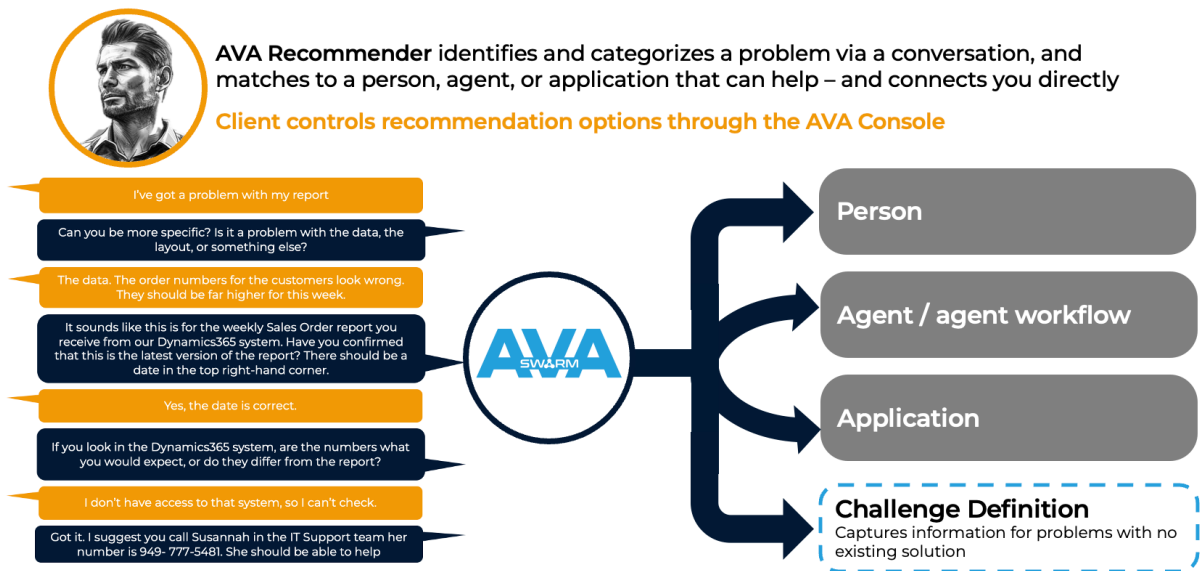
Query Data
[{"from_table": "madanzo.Global", "Total_Volume_Global": 9924043.0}, {"from_table": "madanzo.Benchmark", "Total_Volume_Benchmark": 10029650.877465736}]

2. Recommender: Find Help More Effectively

- **People-based:**
 - I have a problem with the customer delivery to X from Y – who should I speak with to help resolve this?
- **Application-based:**
 - How do I change the demand forecast? Jane does this but she's on vacation.
- **Agent-based:**
 - Can I change the tableau report to filter by regions, and show this as a bar chart instead of a line-graph?



- Note that these are conversations, not a simple Q&A. For example:



3. Automated Alerts

- Requested Alerts:**
 - Let me know if our stock level for X drops below Y in this region.
 - Can you tell me when truck Z arrives at the warehouse?
 - Warn me immediately if we see any new USDA quality reports for Texas
- Proactive Alerts:**
 - We've had 3x more customer rejections this month than usual – would you like to see the details?
 - Truck/load scheduling conflict alert: two transloads are booked for the same bay at 3pm tomorrow.
 - Inventory balancing suggestion: should we transfer stock between facilities based on current usage and deliveries?
 - Cycle time has increased 12% – likely due to loading bay delays and slower rail turnarounds.

4. Support Decisions and Risk Management

- Can you reprice our open wheat offers based on competitor movements this morning?
- Let me know if corn basis has widened in region Y – this might be an arbitrage opportunity.
- Can you trace the rejected load of sorghum to any other loads from the same source?
- Lot 18294 was processed without a certificate of origin – can you flag this for QA?



5. Automating Workflows and Communication

- Can you summarize today's logistics call and action points?
- Several delivery records are missing weights – can you request them from the relevant warehouse teams?
- Dryer throughput is lagging 15% behind intake at the west plant. Can you generate a report that explains why, and shows historical comparisons?

6. Learning & Onboarding

- As a new transload coordinator, here are the five reports and dashboards most relevant to your role.
- What does “basis” mean again? Can you explain it with an example?
- Explain this report to me – what is a lever and why does it matter?

7. Interactive Report Generation

- Can you produce a management summary of last week's trades?
- Please generate a report summarizing our performance month by month for this fiscal year including a historical comparison with 2024. I will write the summary myself, so leave a blank space.
- Let's work together on a challenge definition for the logistics challenge. I'd like to start with the constraints.

The screenshot shows the Madanzo AI interface. On the left, there's a sidebar with 'CONVERSATIONS' and 'DOCUMENTS'. The main area shows a chat with 'AVA'. The chat history includes a conversation about constraints. The current message from the user is: 'Actually, while this is top of mind, can we capture some constraints?'. The AI response is: 'Of course. Here are the typical constraints we see for outbound logistics. Are there any here that don't apply in your scenario? Or other constraints you want to add?'. Below the chat, there's a 'Constraints' table.

Type	Metrics / Dimension	Operator	Limit	Notes
Guideline	Demand variability	=		
Hard Constraint	Safety stock per product	=		
Hard Constraint	Transportation costs	<=		
Hard Constraint	Supplier lead times	=		
Hard Constraint	Regulatory/compliance	=		
Hard Constraint	Location capacities	=		
Hard Constraint	Stockout penalties	=		
Hard Constraint	Storage costs	=		
Hard Constraint	Seasonal requirements	=		
Scope	Facilities	=		
Soft Constraint	Minimum stock level per product	=		

Below the table, there's a text block explaining constraints: 'Constraints are the limits placed on a process, and the bounds within which a solution must be defined. Many constraints are relatively stable throughout business operation, although some may be changed by users at run time, or by conditions (e.g., weather). Some constraints are: mechanical or hardware limitation (e.g., the maximum number of pallets per vehicle), due to law or labor policy (e.g., the maximum vehicle operation hours), linked to external factors (e.g., pickup or delivery time windows at warehouse, destination, etc.), based on current best practice (e.g., keep certain transport lanes separated, etc.). We use the following categories: Hard constraint: something you cannot change – it is fixed by law, physics, or third parties, e.g., speed limit. Soft constraint: a parameter we can change, e.g., the maximum number of miles a truck can drive per day. Scope: is a filter on the data we will use for this optimization, e.g., do not include international orders. Guideline: is a recommendation to help guide the decision process, but can be ignored/broken, if necessary, e.g., use broker x for CA deliveries.'

8. Role-based Intelligence

- **Challenge Engineer:** can interview relevant stakeholders and generate a Challenge Definition report, using her knowledge of the specific function to assist
- **Data Scientist:** can create correlations and aggregations of data and look for insights, either as a one-off, or as an ongoing process
- **Data Cleaner:** can help prepare data for review, removing duplicates and identifying formatting issues, or can generate dummy data
- **Recruiter:** perform initial screening interviews for potential hires, position your company in the correct way, and gather information in a consistent report

9. Organizational Health

- **Surveys:** AVA can interview a single person or perform a group survey, via a template that structures and guides the questions. The information is then used to generate a professional document from a matching template. Data can be summarized, visualized, or reported verbatim depending on the template guidance

10. Strategic Foresight and Planning

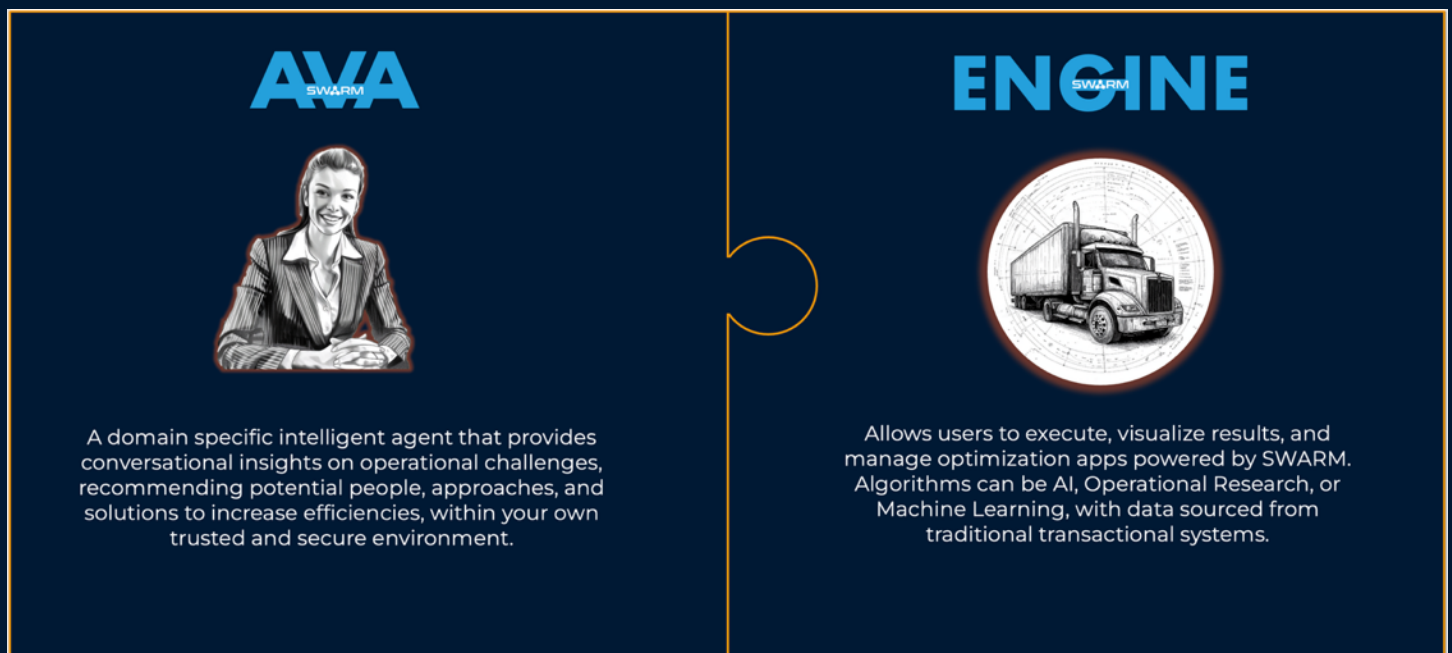
- **Scenario planning:** based on current inventory, weather forecasts, and market trends, what are the potential supply chain risks for the next quarter?
- **What-if modeling:** if we expand our rail capacity by 20%, what impact will that have on throughput and delivery times?
- **Trend identification:** what long-term changes are we seeing in demand patterns for sorghum across our regions?
- **Annual planning:** can you assist in creating a first draft of our operational plan for 2026, using data from this year and last?



Better together

While SWARM AVA and the SWARM ENGINE can each be deployed independently – delivering immediate value in operational insight or advanced optimization – their real power emerges when combined. The two were designed to work seamlessly together.

SWARM AVA identifies challenges, uncovers hidden inefficiencies, defines and recommends solutions. The SWARM ENGINE executes those applications at scale, applying powerful algorithms to optimize outcomes. Together, they form a continuous intelligence loop: discovery, decision, action – all in one secure environment:



Our expert data science and professional services teams work alongside clients to tailor each deployment, accelerate onboarding, and maximize impact. From designing custom models, selecting and refining algorithms, to integrating with complex legacy systems, we help organizations unlock the full power of SWARM AVA and the SWARM ENGINE, turning powerful technology into measurable business outcomes with high value returns.

From insight to impact, faster. Delivering value where it is most needed.

AI-powered optimization for operations.





For more information please contact:

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